COSTVISION · OFFLINE 2D DRAWING INTELLIGENCE

Independent Technical Validation Report

2D engineering drawing reader integrated with 3D CAD should-costing

Readiness: Suitable for controlled internal testing

Date	10 November 2026
Code validated	e8328c23 (measured); final fixes 90022108 / 1166766a (branch claude/new-session-ts4byp)
Test material	26 genuine public CAD + drawing pairs, 3 drawing-only sets; 44 drawings with independent truth files (1700 truth items)
Real prices compared	None — no purchase price, quote or cycle sheet exists for any tested part. Every cost figure here is INDICATIVE.
AI used at runtime	None on the drawing path; the application ran with AI switched off (AIR_GAPPED=1, no key) for every test.

Prepared as an independent audit: code review by a reviewer who had not built the feature; truth files written from rendered drawing images by reviewers who never ran the reader; measurement and reliability runs scripted and repeatable. Evidence files and scripts are listed in the appendix.

1. Executive summary

> **Overall:** The offline drawing reader is safe to put in front of engineers: no drawing leaves the machine, nothing it reads reaches a cost until an engineer confirms it, and CAD-only costing is unchanged. It is not yet an accurate reader. On 44 genuine public drawings it reads 30 % of what they state (58 % on CAD-exported PDFs with selectable text, 22 % on text drawn as curves, 5 % on scans); 69 % of what it reads is right (85 % on selectable-text PDFs). Material is now read on 32 of 37 drawings. Cost accuracy is NOT validated: no real price exists for any tested part.

- Data protection — PASS after fix. The audit found one Critical defect: in AI-led or Compare mode a PDF drawing was sent to the AI model while the screen said nothing leaves the server. Fixed: no drawing is sent to a model in any mode, and a model's reading of a drawing is discarded. Proven by tests with every outbound socket blocked.
- Unconfirmed values never reach a cost — PASS. Across every costing run, an unreviewed drawing costs exactly what no drawing costs (£11.033029 v £11.033029); confirmed drawing answers cost exactly what the same answers typed by hand cost.
- CAD-only regression — PASS. 8 real parts on 8 commodities are bit-identical to the build before the drawing feature; the 40-part real-parts baseline shows no drift; the full suite passes (5,913 tests passed (362 files)).
- Extraction is selective, not complete. Strict precision / recall after the fixes: 69.1 % / 30.1 % overall (83.3 % / 26.6 % before). Reliable: material (100 % precision, 86.5 % recall), general tolerance, coating, notes, fits (6 of 7 right). Unreliable: title block (17 % recall), GD&T (4.7 % recall — symbols drawn as geometry are not read), dimensions on curves and scans, roughness.
- The precision fall is one drawing family: NX stroke-font sheets now go through OCR (they read nothing before) and give 139 wrong dimension readings against 35 right. Every one is marked 'needs checking' and none can reach a cost, but the engineer has to reject them.
- Reconciliation: 12 pass / 6 partial / 2 fail of 20 tests (6 / 8 / 5 before). Conflicts are raised and never settled for the engineer; corrections persist and are traced (who, when, why); a review sent with another drawing is skipped. Not detected: a drawing paired with the wrong CAD when no holes are read, and a revision mismatch (revisions are never compared).
- Costing effect (indicative): where the drawing's material was confirmed, the cost moved by +17 % to +58 % on 7 of 13 rows — largely because the CAD-only run had assumed the wrong material (plastic from a 'PP-' part number in the file name, a defect now fixed). This shows the drawing replacing an assumption, not that either figure is accurate.
- Exports — PASS after fixes. Screen = core PDF = audit pack = Excel to the penny on every costing checked (£15.20, £17.80, £24.92); drawing table matches the review panel row for row and now prints who decided, when and why. Fixed on the way: two different confidence grades for one costing, stale 'no drawing is read' text, a clipped stamp, a dropped depth symbol.
- 52 defects were found by three independent runs; 32 fixed with a test that failed first, 5 partly fixed, 15 open (none Critical or High; 8 Medium, 12 Low). The most consequential open items: machining costs the material FAMILY only (304 stainless is costed as EN8 carbon steel); blank development blocks the server for about 40 s (every user waits); revision and wrong-CAD pairing are not checked.

2. Scope and method

- Independence: the code audit was done by a reviewer who had not built the feature, against the original brief; the 44 truth files were written from rendered drawing images by two reviewers who never ran the reader and never saw its code; measurement and reliability runs are scripts that can be re-run.
- Test material: 26 genuine public CAD + drawing pairs and 3 drawing-only sets, every one with its licence and the evidence that drawing and model are the same part (register in the workbook). Sources: NIST MBE PMI test cases and D2MI parts (public domain), LulzBot TAZ 6 (CC BY-SA 4.0), OpenEAI robot arm (BSD-3), SeedMint jig (GPL-3.0), the DOE/NASA Automotive

Stirling Engine drawings (public use). The project's own generated drawings were NOT counted as evidence.

- Not accessible: GrabCAD, TraceParts, PARTcommunity and 3D ContentCentral need an account; NHTSA blocks automated access; Bosch Motorsport and catalogue vendors' terms forbid it. No login wall, bot check or licence was worked around. Consequence: the only automotive drawings are the Stirling-engine scans.
- Three workflows on identical inputs: A — CAD only (blocking questions answered with the tool's own leaning, recorded); B — drawing only (the reader, scored against truth); C — CAD + drawing reviewed as a careful engineer would from the truth file (confirm right readings, correct wrong ones, reject fabricated ones). UK, 10,000 a year, 5 years, DAP, the built-in rate book, 10 Nov 2026, AI switched off throughout.
- Scoring: precision = correct ÷ (correct + wrong); recall = correct ÷ items stated on the drawing; a reading in a group the truth marks complete that is not on the drawing counts as wrong (fabricated); a missing value is never counted correct. Lenient scoring differs from strict only where the truth writer and the reader file the same diameter under two names (2 readings); no value is relaxed.
- Fix and re-test in three rounds; every fix has a test that failed before it, and after each round the full suite, the real-parts baseline and a live browser run with AI off were repeated. The reader fixes were developed on half the drawing families and checked on the other half (held-out material recall 79 %, 15 of 19).

3. Architecture and requirements compliance

63 requirements from the original brief were audited in the code (file:line evidence in the workbook). Verdicts at the start of the audit and after the defect fixes:

Verdict	At audit	After fixes
Pass	41	50
Partial	20	12
Fail	2	1
Not tested	0	0

Requirements not fully met after the fixes

ID	Requirement	Verdict	Why
C12	DWG via local conversion or licensed solution	Fail	Documented limitation, no converter.
C15	Drawing quality assessment	Partial	
C18	Drawing view and section identification	Partial	
C21	GD&T symbol interpretation where feasible	Partial	F12. Not costed.
C24	Welding and joining specification extraction	Partial	None reaches a cost (listed not costed).
C26	Geometric feature recognition	Partial	
C33	Detect conflicts; never silently overwrite	Partial	F02 fixed; open: the file-name family is not a conflict source (F15, Low); revision and wrong-model pairing not detected (M-D06, M-D07).
C35	Every parameter keeps name, value, unit, source, file, sheet/location, method, status, reliability, override history	Partial	F05 fixed (server stamps who / when); open: a client-supplied drawing file name can relabel the source (F16, Low).
C43	Commodity-independent core with configurable commodity rule modules	Partial	Only the 15 rule-pack commodities have packs; none for brakes, steering, wheels, electrical, battery, EDU, welded assemblies. Reader core contains gear-specific extraction (extract/gear.ts, layout findGearTable).
C52	Test B: drawing-only extraction without costing	Partial	No UI path for a drawing without a CAD file (F13).
C60	Resource limits: OCR memory and queue	Partial	150 MP cap before decode, PNG decoded row-wise, OCR one at a time (A0 scan 910 → 433 MB); JPEG still decodes full size under the cap and the worker has no resourceLimits.

ID	Requirement	Verdict	Why
C61	End-to-end demonstration on a real automotive drawing and its CAD	Partial	No real customer drawing + model pair.
C63	Drawing specs (material, thickness, tolerances, GD&T, roughness, heat treat, coating, welding, threads, inspection) influence the cost	Partial	By design most drawing specifications are shown and reported but not costed (no cost formula was changed); machining now lists a stated grade as not costed (F10).

4. Commodity and component test coverage

Coverage is by component type, judged from the part; the set is not automotive except for the Stirling-engine scans. Commodities the tool costs but which no genuine public pair covered are listed below; they are NOT validated.

Commodity (judged from the part)	Parts / drawings	Validated how far
machining	nist-ctc-01, nist-ctc-04, nist-ctc-05, nist-ftc-06, nist-ftc-07, nist-ftc-08, nist-ftc-10, nist-d2mi-905, nist-d2mi-906, nist-d2mi-907, nist-d2mi-908, nist-d2mi-904, lulzbot-taz6-bed-plate, lulzbot-taz6-bed-leveling-washer, openeai-arm-jxs-01-02, openeai-arm-jxs-02-03, openeai-arm-jxs-05-01, seedmint-jig, nist-d2mi-903, lulzbot-taz6-8mm-rod	extraction + reconciliation + costing A/C (D2MI, LulzBot plates, OpenEAI, FTC-09/10)
casting	nist-ctc-02, ntrs-automotive-stirling-mod1-drawings	extraction only (NIST CTC-02); no casting with a material stated
sheet metal	nist-ctc-03, nist-ftc-09, lulzbot-taz6-electronics-chassis, lulzbot-taz6-electronics-cover, lulzbot-taz6-corner-bracket	extraction + reconciliation + costing A/C on the sheet route (LulzBot bracket and chassis)
rubber	nist-ftc-11	extraction only (NIST FTC-11 ring)
other	nist-am-artifact	extraction only

Not covered at all: suspension, axles and driveline; braking and steering; wheels and tyres; forgings; welded assemblies; plastic mouldings; electrical and electronic parts; battery, EDU and transmission parts. No genuine public CAD + drawing pair was found for these without an account or a licence restriction. The tool's 15 CAD routes were regression-tested CAD-only on the project's own 40 real parts, which is not a drawing validation.

5. Drawing extraction accuracy

Scored against 44 independent truth files. **Precision** = correct ÷ (correct + wrong); **recall** = correct ÷ items the drawing states. A missing value is never counted as correct; a reading in a group the truth marks complete that is not on the drawing counts as wrong (fabricated). Strict scoring is the scorer as written; lenient differs only on two naming equivalences (listed in the workbook).

Before and after the fixes (strict)

Slice	Precision before	Recall before	Precision after	Recall after
All drawings	83.3 %	26.6 %	69.1 %	30.1 %
Class: curves	82.2 %	18.1 %	57.4 %	22.3 %
Class: scan	100.0 %	4.4 %	100.0 %	5.0 %
Class: vector	83.7 %	55.8 %	84.6 %	58.2 %
Family: LulzBot TAZ 6	88.3 %	43.6 %	88.8 %	45.9 %
Family: NIST AM artifact	100.0 %	9.2 %	100.0 %	9.2 %
Family: NIST PMI test cases (CTC/FTC)	75.2 %	18.7 %	76.6 %	19.7 %
Family: NIST D2MI	–	0.0 %	20.1 %	10.0 %
Family: OpenEAI arm (GB)	71.7 %	46.2 %	72.6 %	48.3 %

Slice	Precision before	Recall before	Precision after	Recall after
Family: SeedMint jig	91.7 %	80.8 %	92.1 %	84.8 %
Family: NTRS Stirling scans	100.0 %	1.1 %	100.0 %	2.1 %

By field group, after the fixes

Group	Found	Correct	Wrong	Missed	Precision	Recall
title block	37	37	0	182	100.0 %	16.9 %
material	32	32	0	5	100.0 %	86.5 %
general tolerance	16	16	0	11	100.0 %	59.3 %
roughness	0	0	0	12	–	0.0 %
coating	16	16	0	3	100.0 %	84.2 %
threads	13	9	4	14	69.2 %	39.1 %
fits	7	6	1	1	85.7 %	85.7 %
dimensions	504	281	214	455	56.8 %	37.7 %
GD&T	73	19	6	338	76.0 %	4.7 %
notes	81	80	1	103	98.8 %	43.7 %
welds	1	0	1	0	0.0 %	–
quality	17	15	2	8	88.2 %	65.2 %

What improved: material (0 → 32 of 37), coating (53 → 84 % recall), quality marks (39 → 65 %). What did not: title block (17 % recall — most of these sheets have no conventional title block, and curve-text cells are not read), GD&T (5 % — symbols drawn as geometry), threads (39 %), dimensions (38 % recall, 57 % precision after stroke-font OCR). A reading's status is the safeguard: every OCR reading needs checking and nothing reaches a cost unconfirmed.

Reading time and memory (one file per process, after the fixes): vector median 1.5 s (max 2.2 s), peak 531 MB; curves median 9.6 s (max 26.9 s), peak 947 MB; scan median 12.3 s (max 26.7 s), peak 777 MB.

6. CAD and drawing reconciliation

Before the fixes: 6 PASS, 8 PARTIAL, 5 FAIL, 0 ERROR. After: 12 PASS, 6 PARTIAL, 2 FAIL, 0 ERROR, 0 N/A.

Test	Scenario	Status	Result
pair:nist-d2mi-904	genuine pair: what the consolidated data says (matching, conflicts, missing, not costed, units)	PARTIAL	49 drawing readings; 0 agree with CAD; 0 hole/thread callouts linked to model features; 0 callouts with no matching model feature; 0 conflicts (); missing: none; not costed: 49; units title [] dims ["mm"]
pair:nist-d2mi-906	genuine pair: what the consolidated data says (matching, conflicts, missing, not costed, units)	PASS	51 drawing readings; 0 agree with CAD; 0 hole/thread callouts linked to model features; 0 callouts with no matching model feature; 0 conflicts (); missing: none; not costed: 50; units title [] dims ["mm"]

Test	Scenario	Status	Result
pair:lulzbot-taz6-bed-plate	genuine pair: what the consolidated data says (matching, conflicts, missing, not costed, units)	PASS	39 drawing readings; 1 agree with CAD; 1 hole/thread callouts linked to model features; 0 callouts with no matching model feature; 1 conflicts (title.partName); missing: none; not costed: 30; units title ["mm"] dims ["mm"]
pair:lulzbot-taz6-corner-bracket	genuine pair: what the consolidated data says (matching, conflicts, missing, not costed, units)	PASS	23 drawing readings; 2 agree with CAD; 0 hole/thread callouts linked to model features; 0 callouts with no matching model feature; 0 conflicts (); missing: none; not costed: 21; units title ["mm"] dims ["mm"]
pair:lulzbot-taz6-electronics-cover	genuine pair: what the consolidated data says (matching, conflicts, missing, not costed, units)	PASS	46 drawing readings; 0 agree with CAD; 1 hole/thread callouts linked to model features; 0 callouts with no matching model feature; 2 conflicts (hole.diameter, title.partName); missing: none; not costed: 39; units title ["mm"] dims ["mm"]
pair:openeai-arm-jxs-01-02	genuine pair: what the consolidated data says (matching, conflicts, missing, not costed, units)	PARTIAL	34 drawing readings; 0 agree with CAD; 0 hole/thread callouts linked to model features; 0 callouts with no matching model feature; 0 conflicts (); missing: none; not costed: 34; units title [] dims ["mm"]
pair:seedmint-jig	genuine pair: what the consolidated data says (matching, conflicts, missing, not costed, units)	PARTIAL	33 drawing readings; 0 agree with CAD; 4 hole/thread callouts linked to model features; 0 callouts with no matching model feature; 1 conflicts (feature.count.threads); missing: material.grade; not costed: 26; units title [] dims ["mm"]
pair:nist-ftc-09	genuine pair: what the consolidated data says (matching, conflicts, missing, not costed, units)	PASS	14 drawing readings; 0 agree with CAD; 0 hole/thread callouts linked to model features; 0 callouts with no matching model feature; 0 conflicts (); missing: none; not costed: 11; units title ["in"] dims ["mm"]
pair:nist-ftc-10	genuine pair: what the consolidated data says (matching, conflicts, missing, not costed, units)	PARTIAL	62 drawing readings; 0 agree with CAD; 5 hole/thread callouts linked to model features; 0 callouts with no matching model feature; 1 conflicts (hole.diameter); missing: material.designation, tolerance.generalClass; not costed: 60; units title ["mm"] dims ["mm"]
pair:nist-ftc-07	genuine pair: what the consolidated data says (matching, conflicts, missing, not costed, units)	PASS	46 drawing readings; 0 agree with CAD; 11 hole/thread callouts linked to model features; 0 callouts with no matching model feature; 0 conflicts (); missing: material.designation, tolerance.generalClass; not costed: 44; units title ["in"] dims ["mm"]

Test	Scenario	Status	Result
wrong-cad:bedplate-drawing+d2mi906	(a) a drawing paired with the WRONG CAD model	PASS	wrong model: 2 conflicts [feature.count.threads, title.partName], 2 callouts not in model, 0 linked; identity conflict true; envelope/thickness conflict false. RIGHT model: 1 conflicts, 0 not in model, 1 linked. A pairing question/block: false
wrong-cad:ftc10-drawing+ctc01	(a) a drawing paired with the WRONG CAD model	PARTIAL	wrong model: 1 conflicts [feature.count.holes], 12 callouts not in model, 0 linked; identity conflict false; envelope/thickness conflict false. RIGHT model: 1 conflicts, 0 not in model, 5 linked. A pairing question/block: false
wrong-cad:ftc09-drawing+ftc10	(a) a drawing paired with the WRONG CAD model	PARTIAL	wrong model: 1 conflicts [material.stockThickness], 0 callouts not in model, 0 linked; identity conflict false; envelope/thickness conflict false. RIGHT model: 0 conflicts, 0 not in model, 0 linked. A pairing question/block: false
wrong-cad:jxs-01-02-drawing+jxs-05-01	(a) a drawing paired with the WRONG CAD model	FAIL	wrong model: 0 conflicts [], 0 callouts not in model, 0 linked; identity conflict false; envelope/thickness conflict false. RIGHT model: 0 conflicts, 0 not in model, 0 linked. A pairing question/block: false
wrong-gear-drawing	(b) the existing wrong gear drawing (z 40 against 38 measured)	PASS	teeth conflict: ["cad: 38 ","drawing: 40 "] resolved=false suggested=cad:gear.teeth; cost no drawing £18.64867987419291, with drawing unsettled £18.64867987419291, settled to 38 £18.64867987419291
missing-material:seedmint-slider	(c) a drawing that states no material (truth: blank)	PASS	drawing-only: status 200, material readings [], missing ["part.volume","material.designation"]; with CAD: missing ["material.designation"]; material questions identical to no-drawing: true; answers from drawing {}
revision-mismatch	(d) revision of the model v the drawing	FAIL	the revision is NOT compared: no CAD-side revision parameter on any case; drawing revisions read: [["seedmint base",["drawing: V10"]],["seedmint slider",["drawing: V22"]],["D2MI 904",[]],["LulzBot washer",[]]]
correction-persists	(e) a correction and confirmations sent as drawingReview; /analyze again, then /reanalyze	PASS	/analyze: corrected {"status":"corrected","value":"Powder coat all over, Cardinal C241-BK109 (QA correction)"}, confirmed 4/4, skipped 0; /reanalyze: corrected {"status":"corrected","value":"Powder coat all over, Cardinal C241-BK109 (QA correction)"}, confirmed 4/4, skipped 0; stamped by server: true (qa@test @ 2026-11-10T11:34:33.181Z); drawing answers {"material.family":"aluminium"}
review-on-other-drawing	(e) the same review sent with a DIFFERENT drawing	PASS	/analyze (drawing status in that response: ready) skipped 5; /reanalyze skipped 5/5; first reason: "the decision was taken on another drawing (008f1486), not this one (450c6b3a)"; decided readings on the other drawing: 0; answers {}
correction-supersedes-family	(e) correct the material designation "5052 H32 AL" to "S235JR"	PASS	before: [{"id":"drawing:material.designation:1>material.family","value":"aluminium","status":"needs-validation","source":"derived","raw":"MATERIAL: 5052 H32 AL, 2.03 MM THICK"}]; after: [{"id":"drawing:material.designation:1>material.family","value":"steel","status":"needs-validation"}]; leaning for material.family: ["steel"]

7. CAD-only versus CAD + drawing costing

> **Indicative, not validated.** NOT VALIDATED — no real price exists for any of these parts (scripts/actuals/cad-actuals-input.csv holds no values). Every figure is the tool's own indicative should-cost; C is not claimed to be more accurate than A — only which inputs are stated rather than assumed. Conditions: UK, 10,000 a year, 5 years, DAP (default terms), rate book at HEAD e8328c23, 10 Nov 2026; AI off (AIR_GAPPED=1, no key); headless cost = /api/cad/compare-routes reference row (the server's costMeasuredPart with the same answers, drawing hash and review)

Part	Route	A: CAD only	C: CAD + reviewed drawing	Δ	Δ %	What changed and why
nist-d2mi-904	machining	£7.96	£7.96	£0.00	0.0 %	no answer differs between A and C
nist-d2mi-906	machining	£5.12	£7.83	£2.71	52.8 %	material.family: A plastic -> C aluminium (from drawing (confirmed, "MATERTAL: ALUMINUM, PLATE, 6061-T6, IN ACCORDANCE WITH ASTM B205.")); material-bucket line 'tool wear (inserts, end mills, drills)': A £0.0729 -> C £0.1678 — Buckets moved: material +£1.1386, process +£0.3338, labour +£0.7595, packaging +£0.0020, logistics +£0.0030, overhead +£0.2679, margin +£0.2003
lulzbot-taz6-bed-plate	machining	£17.30	£24.92	£7.63	44.1 %	material.family: A (not answered — costed default / leaning) -> C aluminium (from drawing (confirmed, "Material: 5052 H-32 Aluminum")); material-bucket line 'tool wear (inserts, end mills, drills)': A £0.3971 -> C £0.9661 — Buckets moved: material +£4.3453, process +£0.8089, labour +£1.0881, packaging +£0.0200, logistics +£0.0500, overhead +£0.7490, margin +£0.5649
lulzbot-taz6-corner-bracket	machining	£3.50	£5.51	£2.01	57.5 %	material.family: A (not answered — costed default / leaning) -> C aluminium (from drawing (confirmed, "Material: 5052 H-32 Aluminum")); material-bucket line 'tool wear (inserts, end mills, drills)': A £0.032 -> C £0.0761 — Buckets moved: material +£0.9497, process +£0.3244, labour +£0.3859, logistics +£0.0020, overhead +£0.1993, margin +£0.1489
lulzbot-taz6-electronics-cover	machining	£11.99	£11.99	£0.00	0.0 %	no answer differs between A and C
lulzbot-taz6-electronics-chassis	machining	refused	–	–	–	material.family: A plastic -> C aluminium (from drawing (confirmed, "MATERIAL: 5052 H32 AL, 2.03 MM THICK"))
openeai-arm-jxs-01-02	machining	£9.14	£13.28	£4.14	45.3 %	material.family: A plastic -> C aluminium (from drawing (confirmed, "6061铝 1件")); material-bucket line 'tool wear (inserts, end mills, drills)': A £0.1259 -> C £0.2988 — Buckets moved: material +£2.9181, process +£0.1940, labour +£0.2976, packaging +£0.0100, logistics +£0.0050, overhead +£0.4092, margin +£0.3067
nist-ftc-09	machining	£10.34	£16.29	£5.95	57.5 %	material.family: A plastic -> C steel (from drawing (confirmed, "MATERIAL: (.1195 THICK) 11 GA 304 SST SHT, ASTM A240. 08")); material-bucket line 'tool wear (inserts, end mills, drills)': A £0.2401 -> C £1.7743 — Buckets moved: material +£2.7288, process +£0.7534, labour +£1.3237, packaging +£0.0420, logistics +£0.0810, overhead +£0.5767, margin +£0.4405
nist-ftc-10	Gravity die casting + machining	£14.39	£14.39	£0.00	0.0 %	no answer differs between A and C
lulzbot-taz6-corner-bracket@sheet	sheet metal fab	£0.50	£0.65	£0.15	29.7 %	material.family: A steel -> C aluminium (from drawing (confirmed, "Material: 5052 H-32 Aluminum")); service.finish: A (not answered — costed default / leaning) -> C powder_coat (from drawing (confirmed, "Break all sharp edges prior to powder coat")); material-bucket line 'consumables (not itemised by the route's module)': A £0.0087 -> C £0.0605 — Buckets moved: material +£0.088
lulzbot-taz6-electronics-cover@sheet	sheet metal fab	refused	–	–	–	service.finish: A (not answered — costed default / leaning) -> C powder_coat (from drawing (confirmed, "A Ad Part is to be powder coated A with Cardinal color C241-BK109."))
lulzbot-taz6-electronics-chassis@sheet	sheet metal	£17.02	£19.91	£2.89	17.0 %	material.family: A steel -> C aluminium (from drawing (confirmed, "MATERIAL: 5052 H32 AL, 2.03 MM THICK")); material.grade: A (not answered — costed default / leaning) -> C mat-aa5052 (from drawing (confirmed, "MATERIAL: 5052 H32 AL, 2.03 MM THICK")); service.finish: A (not answered — costed default / leaning) -> C powder_coat (from drawing (confirmed, "BREAK ALL CORNERS PRIOR
nist-ftc-09@sheet	sheet metal	refused	–	–	–	no answer differs between A and C

Every change from A to C is explained by an answer the drawing supplied (material family or grade, finish). No wrong reading reached a cost in any run. Machining asks the material family only, so a stated grade (304, 6061-T6) is costed as the family's representative grade and listed as not costed — a cost-model limitation the drawing exposes. The 'PP-' file-name defect made the CAD-only figure for two LulzBot parts plastic; it is fixed in 90022108 (proven by a unit test on the real file names), so those two A figures will move; they were not re-costed live after the fix.

8. Independent costing accuracy assessment

- Independent costing accuracy cannot be assessed: there is no purchase price, supplier quote or cycle-time sheet for any tested part, and none was invented. Every £ in this report is INDICATIVE.
- What the comparison does show: on 7 of 13 cost rows the drawing replaced an assumed material (and on two a finish) with a stated one; on 3 it changed nothing; 3 were refused by the tool's own route check in both A and C; it never made a cost depend on an unconfirmed reading.
- What it does not show: that the CAD + drawing figure is closer to what a part costs. A difference between A and C is a difference in inputs, not evidence of accuracy.
- To measure accuracy: enter real prices for real parts in calculator/scripts/actuals/cad-actuals-input.csv (price, volume, life, country, finish, source) and run npx tsx scripts/cad-actuals.ts then npm run accuracy. Ten to twenty parts per commodity, with their drawings, would give a first error band per commodity.

9. PDF and Excel export audit

Exports were opened, every PDF page rendered and read (76 pages in the first run), every worksheet range scanned and recalculated in LibreOffice (0 formula errors; the CAD workbook holds values and a checks sheet by design). Four export defects were found and fixed (two confidence grades, stale drawing text, clipped stamp, dropped symbols); the final run on the bed plate with a corrected reading agrees screen = PDF = audit pack = Excel at £24.92 and prints who / when / why.

Check	Before fixes	Expected	Observed
screen = PDF = Excel = headless to the penny: casting bracket + real-casting-bracket.pdf, review confirmed (Confirm all read exactly + tolerance + masks + part number corrected)	pass	one figure to the penny on screen, core PDF, audit pack, workbook and the comparison reference row	screen £15.20; core PDF page 1 £15.20 (text check); audit pack cover £15.20; workbook 1-Summary TOTAL 15.199446 (→ £15.20); compare-routes reference row 15.200090; screen stamp "United Kingdom rate book v2.2.0 · built-in · book 9b7aee68 · rules v9 · kernel 67caebef99ba · drawing reader v1.0.0 · real
screen = PDF = Excel = headless to the penny: gear + gear-drawing-WRONG-z40.pdf, teeth conflict settled to the 3D model	pass	one figure to the penny on screen, core PDF, audit pack, workbook and the comparison reference row	screen £17.80; core PDF page 1 £17.80 (text check); audit pack cover £17.80; workbook 1-Summary TOTAL 17.795588 (→ £17.80); compare-routes reference row 17.795588; screen stamp "United Kingdom rate book v2.2.0 · built-in · book 9b7aee68 · rules v9 · kernel 67caebef99ba · drawing reader v1.0.0 · gear
screen = PDF = Excel = headless to the penny: gear, no drawing	pass	one figure to the penny on screen, core PDF, audit pack, workbook and the comparison reference row	screen £17.80; core PDF page 1 £17.80 (text check); audit pack cover £17.80; workbook 1-Summary TOTAL 17.795588 (→ £17.80); compare-routes reference row 17.795588; screen stamp "United Kingdom rate book v2.2.0 · built-in · book 9b7aee68 · rules v9 · kernel 67caebef99ba · 10 Nov 2026"; page errors []
every PDF page rendered (pdftoppm 70 dpi + zooms) and looked at; text extracted; blank pages, replacement glyphs, NaN/undefined/null scanned	fail	no blank page, no broken glyph, no clipped or overlapping text, symbols Ø ± ° µm render	0 blank pages, 0 replacement / NaN / undefined strings in any PDF; Ø, ±, °, µm render in every report (counts in inspection.json). Found: audit §1 reproducibility stamp clipped at the page edge with a drawing (D04); ↧ dropped from the thread value in §9c (D05); §9b heading overlap on the casting aud

Check	Before fixes	Expected	Observed
every worksheet opened with openpyxl, every populated cell scanned; recalculated with LibreOffice headless and cached v recalculated values compared	fail	no formula errors, no broken references, no hidden sheets / rows, no NaN / undefined text; money numeric; drawing workbooks carry 10-Drawing, the no-drawing workbook does not	0 formula errors after recalculation; the CAD workbooks hold NO live formulas (cells beginning "= …" are text labels such as "= should-cost", stored as strings), so recalculation cannot disagree; 0 hidden sheets, rows or columns; 0 suspicious strings. 10-Drawing present on both drawing workbooks and
workbook 10-Drawing rows = the review panel rows (label, value, status), and the conflict outcome	pass	same rows, values and status words; conflict outcome as settled on the panel	casting bracket: 24 + 20 not-costed rows on the panel = 24 + 20 in 10-Drawing, 0 differences in label/value/status. Gear: 9 + 3 = 9 + 3, 0 differences; conflict row "Number of teeth | 3D model: 38 v Drawing: 40 | 3D model (38) | Settled by QA Engineer: 38 (the 3D model)" = panel "Settled by QA Engin
one confidence grade across screen, core PDF, audit pack and workbook	fail	same grade everywhere	screen / core / 0-Executive: Medium; audit pack cover, executive summary, §7 and 1-Summary: Low — on all three costings, including the one with no drawing (D06).
report text consistent with a drawing being attached	fail	no statement that no drawing is read; thread note consistent with the drawing	core §7 and 0-Executive say "(no drawing is read)" (D03); core page 1 says the CAD model shows no thread while the drawing states M6x1-6H (D13).
no-drawing costing reports carry no drawing content	pass	no §9c, no 10-Drawing, stamp without reader	no "Drawing specification" / "§9c" / "drawing reader" text in core or audit; workbook sheets 0-Executive … 9-DFM (no 10-Drawing); stamp "United Kingdom rate book v2.2.0 · built-in · book 9b7aee68 · rules v9 · kernel 67caebef99ba · 10 Nov 2026". Audit pack text equals the wrong-drawing gear audit pac
GD&T symbols (⌖ ⊥ Ⓜ) in the PDF / workbook drawing tables	not-tested	-	none of the three drawings exported carries a GD&T frame (the generated gen-* sheets do, but no CAD part is paired with them), so the rendering of these glyphs in reports was not exercised

10. Functional and regression testing

Reliability and negative run (110 scripted cases, before the fixes): 86 pass, 13 fail, 6 partial, 5 not-tested. Every failure became a defect in section 11.

- Full suite after each fix round: 5,883 → 5,912 → 5,913 tests across the three fix rounds, all passing.
- Real-parts baseline (40 real parts re-measured with the geometry kernel): no drift after every round.
- CAD-only, current build v the build before the drawing feature (36f6f7d3), same answers: 8 parts identical to the last decimal on total, all 8 buckets, inputs and questions. The one intended CAD-only change is the round-3 'PP-' / 'PC-' part-number fix, which only affects file names carrying such a code.
- After the final fix (90022108) the LulzBot 'PP-' part numbers no longer read as plastic (unit test on the real file names); the live CAD-only re-cost of those two parts was not repeated, so section 7 keeps their measured (plastic) A figures.
- Live browser run with AI off (5 scenarios, 73 checks) after every round: 73 / 73.
- Offline proof: the drawing path with every outbound connection blocked and AIR_GAPPED=1 — no connection attempted.

Area	Pass	Fail	Partial	Not tested
input	58	4	4	3
reprocess	14	5	2	1
regression	9	0	0	0
export	5	4	0	1

11. Defects, corrective actions and limitations

52 defects across the three runs: 32 fixed, 5 partly fixed, 15 open. By severity: Critical 1, High 4, Medium 22, Low 25. Every fix has a test that failed before it (named in the workbook); the full suite, the real-parts baseline and the live browser run passed after each round.

Not fully fixed

ID	Severity	Status	Defect	Recommended action
A-F08	Medium	Partly fixed	Image and TIFF drawings are decoded at full resolution before any pixel cap (decompression-bomb risk)	Read width × height from the header (PNG IHDR, JPEG SOF, TIFF IFD tags 256/257) and refuse or downsample before decoding; decode TIFF pages one at a time; give the worker resourceLimits or run OCR in a child process.
M-D01	Medium	Partly fixed	Drawings whose annotation is stroke-font polylines (NIST D2MI, Siemens NX) are classed as vector text: no OCR runs, nothing is read, and no warning says so	stroke-font sheets now OCR'd: D2MI recall 0 → 10 %, but 139 wrong (all needs-validation) — see M-D16
M-D06	Medium	Open	Model revision is never compared with the drawing revision	Model revision never compared with the drawing revision
M-D07	Medium	Partly fixed	A drawing paired with the wrong model is not detected unless hole callouts were read; overall size / thickness are never compared	FTC-09 drawing on the FTC-10 model now raises a stock-thickness conflict; OpenEAI pairing still unflagged
M-D09	Medium	Open	(CAD-only, pre-existing) Blank development runs on the server's event loop for ~40 s: the server stops answering, and a client's next request on a kept-alive socket is reset (ECONNRESET)	Pre-existing: blank development blocks the server event loop ~40 s (deployment risk)
M-D14	Medium	Open	The machining route costs the material FAMILY only: a stated grade on the drawing (304 stainless, 6061-T6, 5052-H32) is replaced by the representative grade (EN8 / 080M40 carbon steel, 6082-T6 bar)	
M-D16	Medium	Open	OCR of stroke-font sheets produces many wrong dimension readings (D2MI: 139 wrong v 35 correct)	
R-D12	Medium	Partly fixed	Server memory after OCR reads: node RSS grows to ~1.1-1.2 GB and stays; peak 2.0 GB (node + geometry worker) on an A0 image	Measure in the 2 GB Docker image; cap the OCR image size; terminate the OCR worker after each pass so its memory is returned.
A-F12	Low	Open	GD&T characteristics are not recognised on the public vector drawing documented as using the AIGDT symbol font	Fix README wording or the font detection; add a geometry-based symbol matcher for the 14 characteristics if GD&T is ever costed.
A-F13	Low	Open	Drawing-only extraction exists only as an API and CLI — the application UI cannot read a drawing without a CAD file	Add a 'Read a drawing' entry that calls /api/drawing/read and shows the same panel with no costing, or state in the docs that Test B is API/CLI only.
A-F14	Low	Open	Hole-and-boss count for the mask cap is computed in three places	Export one measuredMaskableFeatures(geo) from the engine and use it everywhere.
A-F15	Low	Open	A confirmed drawing family silently outranks the family read from the CAD file name; no conflict row	Map the file-name family into consolidate() as source 'name' so a disagreement is shown as a conflict with the precedence suggestion.
A-F16	Low	Open	Client-supplied drawingFile name replaces the source file recorded on every parameter	Keep the name recorded at upload as sourceFile; show the request's name only as a label.
A-F19	Low	Open	Company rate-book grades cannot be matched from a drawing	Resolve grades against the active (resolved) book.
M-D10	Low	Open	(CAD-only, pre-existing) Route leaning contradicts the tool's own plausibility guard on sheet parts	Pre-existing: route leaning contradicts the sheet-metal guard
M-D11	Low	Partly fixed	Split hole callouts raise false count conflicts: each callout's count is compared with the model's total for that diameter	FTC-07: 4 false conflicts → 0; FTC-10: one residual (two Ø6 read as dimensions are not summed)

ID	Severity	Status	Defect	Recommended action
M-D13	Low	Open	Shipped accuracy script mislabels a file's content class when the truth's `files` map is keyed by another file name	Scoring script labels NIST copies by file name (test tooling only)
M-D17	Low	Open	On the screen the material DESIGNATION reading has no Confirm button (only its interpreted family does)	
R-D11	Low	Open	OCR structural mis-assignments on skewed / low-resolution scans (wrong title-block cells, Ø read as 9, decimal point lost)	Warn on low resolution (< 150 dpi) and reject title-block values that do not fit their cell (a scale must look like n:m).
R-D15	Low	Open	DXF reader joins the next note's number to a note and does not read a 'PART No.' text	Check line assembly for DXF TEXT that starts with a note number.

Fixed

ID	Severity	Defect	Commit	Proof
A-F01	Critical	With Compare or AI-led mode on, the PDF drawing is still uploaded to Anthropic while the screen says 'Read on this server — no AI, nothing leaves it'	59b0aa70	tests/drawing-validation-fixes.test.ts F01/F02: no drawing document in any model request; model drawing reads stripped
A-F02	High	In AI mode, a second, unconfirmed drawing reading (the model's) reaches the cost beside the offline review	59b0aa70	same test: stripModelDrawingReads drops finish and gear drawing fields
M-D03	High	Material designation read on 0 of 24 genuine drawings that state one (recall 0/37 material facts); the drawing never answers the material question on any genuine pair	2a4b54c2	drawing-reader-gaps: material grammar (0 → 94 % / 79 % recall) — confirmed in the final re-measurement
M-D08	High	(CAD-only, pre-existing) A part-number prefix 'PP-' is read as polypropylene and the material family is pre-answered 'plastic' and shown as ANSWERED	90022108	drawing-validation-fixes D08: the prefix is matched anywhere in the name — the real LulzBot names (…_PP-MP0108) did not move in the re-measurement; fixed in 90022108
R-D01	High	The word GEOMETRIC is read as the zinc-flake trade name GEOMET: a note about geometric tolerances becomes an exact-text coating reading that "Confirm all read exactly" applies	59b0aa70	drawing-validation-fixes D01: GEOMETRIC/GEOMETRY read no coating
A-F03	Medium	Values worked out from an OCR reading are marked 'extracted' (no doubt) while their OCR source stays 'needs-validation'	59b0aa70	drawing-validation-fixes F03: OCR-derived values are needs-validation
A-F04	Medium	A saved review is replayed by parameter id only; a reader change can make a 'confirm' land on a different reading	59b0aa70	drawing-review-binding.test.ts (12): decisions bound to drawing hash and seen value
A-F05	Medium	Override history 'by' and 'at' are whatever the client sends — not the authenticated user	59b0aa70	drawing-review-binding / drawing-integration-routes: server stamps who and when
A-F06	Medium	Drawings, readings, sheet pictures and OCR jobs are not scoped to the user who uploaded them	59b0aa70	drawing-access-scope.test.ts (7): two users and an admin
A-F07	Medium	Drawing files and readings outlive their stated retention on a long-running server	59b0aa70	drawing-access-scope: hourly sweeper
A-F09	Medium	'AlSi7Mg0.3' (A356 / LM25) is offered only A357 (AlSi7Mg0.6) as its grade candidate	59b0aa70	drawing-validation-fixes F09: EN 1706 designations
A-F10	Medium	On machining, a drawing grade (e.g. 42CrMo4) only answers the family; the grade itself is neither costed nor listed as 'not costed'	59b0aa70	drawing-validation-fixes F10: grade listed as not costed on machining
A-F11	Medium	ISO 2768-m (the default title-block note) maps to tolerance class 'tight', which on ferrous castings means investment casting	59b0aa70	engineering-data-interpret: ISO 2768 / DIN 7168 answer no as-cast class
M-D02	Medium	Scanned sheets carrying a scanner OCR text layer are read as EXACT text when the page has more than 20 path operators	2a4b54c2	drawing-reader-gaps: invisible / under-image text layer re-OCR'd — confirmed in the final re-measurement

ID	Severity	Defect	Commit	Proof
M-D04	Medium	Correcting a misread material designation does not re-interpret the material family the reader derived from it: a stale 'steel' stays on the panel as a leaning for the material question	2a4b54c2	drawing-validation-fixes D04: superseded family on correction — confirmed in the final re-measurement
M-D05	Medium	Reports do not say who confirmed or corrected a reading, or when; the correction reason is not printed	2a4b54c2	drawing-validation-fixes D05: who / when / why in reports — confirmed in the final re-measurement
R-D02	Medium	A review made on one drawing is replayed onto a different drawing: positional reading ids match a different reading with a different value	59b0aa70	drawing-review-replay-route.test.ts: a review on another drawing is skipped
R-D06	Medium	Two confidence grades for one costing: screen, core PDF and 0-Executive say Medium; audit pack and 1-Summary say Low	59b0aa70	report-reliability-fixes + casting-360-review: one grade (costingConfidence) on screen, PDF, workbook
R-D09	Medium	A scan turned 90° is reported as read upright but yields a wrong material (S310N → steel) for an aluminium A356-T6 drawing	59b0aa70	drawing-raster-limits: four-turn orientation probe (A16c reads A356-T6)
A-F17	Low	Stale and misleading text: pack comment says headless does not carry finish fields; log says the drawing is 'attached' on the no-AI path	59b0aa70	comment corrected
A-F18	Low	engineering-data-packs test times out under parallel load	59b0aa70	engineering-data-packs: stated 30 s timeout
M-D12	Low	Reader attaches flag-note numbers to callouts and truncates a leading digit	2a4b54c2	drawing-reader-gaps: flag-note digits — confirmed in the final re-measurement
M-D15	Low	"5052 H-32 Aluminum" maps to no sheet grade ("5052 H32 AL" does): the grade question stays at the default AA5754-H22	90022108	drawing-validation-fixes D15: "5052 H-32 Aluminum" names AA5052-H32 (to confirm)
R-D03	Low	Core PDF and workbook say "no drawing is read" on a costing that has a drawing read and applied	59b0aa70	report-reliability-fixes: drawing-aware wording
R-D04	Low	Audit pack §1: the reproducibility stamp runs off the right edge of the page when a drawing is attached	59b0aa70	report-reliability-fixes: stamp wraps
R-D05	Low	PDF drops the thread-depth symbol: "M6x1-6H ↧ 12" prints as "M6x1-6H 12"	59b0aa70	report-reliability-fixes: drawing symbols as words in the PDF
R-D07	Low	30 MB exactly: /api/drawing/read refuses it (413) while /api/cad/analyze accepts it	59b0aa70	drawing-access-scope: exactly 30 MB accepted on both routes
R-D08	Low	Over-limit message rounds the size: "Drawing is 30 MB; the limit is 30 MB."	59b0aa70	413 message states bytes and the limit
R-D10	Low	Rotation is reported inconsistently: sheet.rotationDeg and the warning disagree	59b0aa70	one rotation convention in stats and warning
R-D13	Low	Core page 1 says the CAD model shows no thread while the confirmed drawing on the same report states M6x1-6H	59b0aa70	report-reliability-fixes: threads stated on the drawing are named
R-D14	Low	Audit pack §9b heading: the subtitle overlaps the title	59b0aa70	report-reliability-fixes: §9b subtitle wraps
R-D16	Low	Review cache key includes each decision's timestamp and author	59b0aa70	drawing-review-binding: review key depends on decisions only

12. Production readiness recommendation

Suitable for controlled internal testing

Controlled internal testing, not yet a Director demonstration of accuracy and not production. For: nothing leaves the machine; nothing unconfirmed reaches a cost; CAD-only costing is unchanged; every reading is traceable and every decision recorded; exports agree to the penny. Against: the reader finds 30 % of what a drawing states (58 % on CAD-exported PDFs); GD&T, title blocks and dimensions on curve-text or scanned sheets are weak; no automotive

CAD + drawing pair could be tested; cost accuracy is unvalidated; four Medium defects remain open that matter in daily use (machining costs the family only, the server blocks for ~40 s during blank development, revision and wrong-CAD pairing unchecked).

Before presenting to the Cost Engineering Director

1. Run the tool's own accuracy script inside JLR on 20–30 real JLR drawings with truth files (scripts/drawings/accuracy.ts) — the only extraction figure that matters, measured without the drawings leaving JLR.
2. Enter purchase prices for 10–20 parts that also have drawings (scripts/actuals/cad-actuals-input.csv) and report the error band, labelled per commodity.
3. Demonstrate on CAD-exported PDFs with selectable text (the class that reads 85 % right), and say plainly that scans and curve-text sheets are read poorly and always need checking.
4. Fix or disclose the open Medium items: grade on the machining route (304 costed as EN8), blank development blocking the server (multi-user deployment), revision and wrong-model checks.
5. Do not present the CAD + drawing cost change as an accuracy gain: it is an input change until real prices are compared.

13. Final independent assessment

1. **1. Does the reader genuinely interpret engineering information, or mainly OCR and text extraction?**
   Both, with a clear line. It extracts text (exact PDF text, or OCR) and then interprets it with standards tables: ISO 286 fits into limits, ISO 8062-3 casting grades and EN 10243 forging grades into the tolerance class, material designations into a family and rate-book grade (EN / AA / EN 1706 / GB / JIS numbering), coating names and thicknesses into the finish routes, hole and thread callouts matched to the 3D model's holes, inch values converted. It does not interpret GD&T symbols drawn as geometry, weld symbols, roughness symbols or views and sections, and it does not decide the process.
2. **2. Which parameters are reliable, which are not?**
   Reliable (high precision; recall as stated): material 100 % / 86.5 %, general tolerance 100 % / 59 %, coating 100 % / 84 %, notes 99 % / 44 %, fits 86 % / 86 % (6 of 7). Unreliable: title block (100 % right but 17 % found), GD&T (4.7 % found), threads (69 % / 39 %), dimensions (57 % / 38 %, much worse on curves and scans), roughness (0 of 12).
3. **3. Does combining 2D drawings with 3D CAD improve engineering-data completeness?**
   Yes, for what the 3D model cannot carry: on the costed pairs the drawing supplied material, temper, finish and coating, which the CAD-only run had to assume (and in two cases assumed wrongly). Measured: 7 of 13 cost rows had an assumed input replaced by a stated one. It adds nothing on drawings whose text it cannot read.
4. **4. Does it demonstrably improve should-costing accuracy against independent reference data?**
   Not demonstrated. There is no independent reference cost for any tested part, so accuracy is unmeasured. The tool's costs changed by +17 % to +58 % where the material was supplied — evidence that the drawing matters, not that the result is more accurate.
5. **5. Which commodities and component types have actually been validated?**
   Drawing extraction: machined aluminium and steel plates and blocks, bent and flat sheet-metal parts, a cast-like housing, a non-rigid ring and scanned engine parts — 44 drawings. Costing A v C: machining and sheet metal only. Not validated with drawings: castings with a stated material, forgings, mouldings, rubber, welded assemblies, brakes, steering, suspension, wheels, electrical, battery, EDU, transmission.
6. **6. Has existing CAD-only functionality remained unaffected?**
   Yes. Bit-identical costs on 8 real parts against the build before the feature, no drift on the 40-part baseline, the full suite passing. The only CAD-only change is deliberate: a 'PP-' / 'PC-' part number in a file name is no longer read as polypropylene / polycarbonate.
7. **7. Are the PDF and Excel outputs technically reliable?**
   Yes after the fixes: screen, core PDF, audit pack and workbook agree to the penny on every costing checked; the drawing table matches the review panel; engineering symbols print (Ø, ±,

°, µm; GD&T and depth symbols as words in the PDF, Unicode in Excel); who / when / why are printed. Four export defects were found and fixed with tests.

8. **8. What must be fixed before presenting to the Cost Engineering Director?**
   Measure extraction on real JLR drawings inside JLR; enter real prices for a sample of parts; fix or disclose the open Medium defects (machining grade, blank development blocking the server, revision / wrong-model checks, stroke-font OCR noise); demonstrate on selectable-text PDFs and state the limits.
9. **9. Ready for an internal trial, a Director demonstration or production?**
   Suitable for controlled internal testing. It is safe and traceable enough for engineers to try on real JLR drawings inside JLR, which is also the step that produces the evidence a Director demonstration needs. Not yet production: accuracy is unmeasured on real drawings and real prices, and four Medium defects remain.

Appendix — evidence

- Plan: validation/TEST-PLAN.md. Code audit: validation/audit/compliance.json (63 requirements), findings.json (19).
- Dataset: validation/dataset/register.json (29 entries with licences), not-used.json (25 sources and why).
- Truth files: validation/truth/*.truth.json (44, written from images, independent of the reader).
- Reliability and export run: validation/reliability/results.json (110 cases), defects.json (16), validation/exports/ (renders).
- Measurement before / after: validation/runs/extraction.json, extraction-final.json, reconciliation(-final).json, costing(-final).json, regression(-final).json, timings(-final).json, exports-final/, raw reader output raw/ and raw-final/.
- Defect register: validation/deliverables/defect-register.json (52). Fixes: commits 59b0aa70, 2a4b54c2 and 90022108 on branch claude/new-session-ts4byp; tests drawing-validation-fixes, report-reliability-fixes, drawing-review-binding, drawing-access-scope, drawing-raster-limits, drawing-review-replay-route, drawing-reader-gaps.